COMP9602: Optimization

Algorithms for Equality Constrained Optimization

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Roadmap

Theory	convex set convex function standard forms of optimization problems, quasi-convex optimization linear program, integer linear program quadratic program geometric program semidefinite program vector optimization duality
Algorithm	unconstrained optimization equality constrained optimization interior-point method localization methods subgradient method decomposition methods

Equality constrained minimization

$$\begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & Ax = b \end{array}$$

- f convex, twice continuously differentiable
- $A \in \mathbf{R}^{p \times n}$ with $\mathbf{rank}\, A = p < n$
- we assume $p^\star$ is finite and attained

optimality conditions: $x^\star$ is optimal iff there exists a $\nu^\star \in \mathbf{R}^p$ such that

$$\nabla f(x^\star) + A^T \nu^\star = 0, \qquad Ax^\star = b$$

(page 10, 4_Convex_Programs_I_COMP9602_Fall2026.pdf; also the KKT conditions)

solving the equality constrained optimization
<=> find the solution to the above KKT conditions (n+p equations)

Examples that are analytically solvable

- quadratic f(x) (with $P \in \mathbf{S}_+^n$)

$$\begin{array}{ll} \text{minimize} & (1/2)x^T Px + q^T x + r \\ \text{subject to} & Ax = b \end{array}$$

Optimality conditions:

$$\begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x^\star \\ \nu^\star \end{bmatrix} = \begin{bmatrix} -q \\ b \end{bmatrix}$$

- coefficient matrix is called KKT matrix

if the KKT matrix is nonsingular, (x^*, ν^*) can be uniquely decided

Solving equality constrained optimization —Method 1

Eliminating equality constraints

represent solution of $\{x \mid Ax = b\}$ as

$$\{x \mid Ax = b\} = \{Fz + \hat{x} \mid z \in \mathbf{R}^{n-p}\}$$

- $\hat{x}$ is (any) particular solution
- range of $F \in \mathbf{R}^{n\times(n-p)}$ is nullspace of A ($\mathbf{rank}\, F = n - p$ and $AF = 0$)

(see also page 12, 4_Convex_Programs_I_COMP9602_Fall2026.pdf)

reduced or eliminated problem

$$\text{minimize} \quad f(Fz + \hat{x})$$

- an unconstrained problem with variable $z \in \mathbf{R}^{n-p}$
- from solution $z^\star$, obtain $x^\star$ and $\nu^\star$ as

$$x^\star = Fz^\star + \hat{x}, \qquad \nu^\star = -(AA^T)^{-1}A\nabla f(x^\star)$$

Example

- optimal allocation with resource constraint

$$\begin{array}{ll} \text{minimize} & f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n) \\ \text{subject to} & x_1 + x_2 + \cdots + x_n = b \end{array}$$

reduced problem:

$$\text{minimize} \quad f_1(x_1) + \cdots + f_{n-1}(x_{n-1}) + f_n(b - x_1 - \cdots - x_{n-1})$$

(variables $x_1, \ldots, x_{n-1}$)

eliminate $x_n = b - x_1 - \cdots - x_{n-1}$, *i.e.*, choose

$$\hat{x} = b \begin{bmatrix} 0 \\ 0 \\ \cdots \\ 1 \end{bmatrix} \qquad F = \begin{bmatrix} I \\ -\mathbf{1}^T \end{bmatrix} \in \mathbf{R}^{n \times (n-1)}$$

Solving equality constrained optimization —Method 2

- Solve the dual, and then recover optimal primal variable x*
  - Lagrangian dual function:

$$\begin{aligned} g(\nu) &= -b^T\nu + \inf_x (f(x) + \nu^T A x) \\ &= -b^T\nu - \sup_x ((-A^T\nu)^T x - f(x)) \\ &= -b^T\nu - f^*(-A^T\nu) \end{aligned}$$

  - The dual problem is

$$\max -b^T\nu - f^*(-A^T\nu)$$

• (possibly) unconstrained optimization: if $g(\nu)$ twice differentiable, descent methods can be applied

• strong duality holds

• reconstruct x*: x* minimizes $L(x, \nu^*)$

Example

$$\min f(x) = -\sum_{i=1}^{n} \log x_i$$

$$\text{s.t. } Ax = b$$

Using

$$f^*(y) = \sum_{i=1}^{n}(-1 - \log(-y_i)) = -n - \sum_{i=1}^{n} \log(-y_i)$$

The dual problem is

$$\max -b^T\nu + n + \sum_{i=1}^{n} \log(A^T\nu)_i$$

After solving the dual, recover primal optimal point:

$$x_i = 1/(A^T\nu)_i$$

Solving equality constrained optimization —Method 3

- Newton's method with equality constraints

$$\begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & Ax = b \end{array}$$

- General idea
  - start with a feasible $x^{(0)},\ \text{s. t. } Ax^{(0)} = b$
  - find a feasible direction (Newton's direction or step):

$$\triangle x_{nt},\ \text{s.t. } x^{(k)} + t \triangle x_{nt} \text{ still satisfies } Ax = b$$

$$<=> A \triangle x_{nt} = 0$$

Newton's direction with equality constraints

- How to find $\triangle x_{nt}$

Δx_{nt} solves second order approximation (with variable v)

$$\begin{array}{ll} \text{minimize} & \widehat{f}(x+v) = f(x) + \nabla f(x)^T v + (1/2) v^T \nabla^2 f(x) v \\ \text{subject to} & A(x+v) = b \end{array}$$

Newton step Δx_{nt} of f at feasible x is given by solution v of

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -\nabla f(x) \\ 0 \end{bmatrix}$$

KKT matrix (see also slide 4)

Newton's step is defined only at points where KKT matrix is nonsingular

When f is nearly quadratic, $x + \triangle x_{nt}$ is a very good estimate of x*, w is a good estimate of the optimal dual variable ν^*

Newton's decrement with equality constraints

- Newton's decrement

$$\lambda(x) = \left(\Delta x_{\mathrm{nt}}^T \nabla^2 f(x) \Delta x_{\mathrm{nt}}\right)^{1/2} = \left(-\nabla f(x)^T \Delta x_{\mathrm{nt}}\right)^{1/2}$$

(The same $\triangle x_{nt}$ -based definition as for unconstrained optimization on page 10, 11_Alg_Unconstrained_Opt_COMP9602_Fall2026.pdf)

properties

- gives an estimate of $f(x) - p^\star$ using quadratic approximation $\widehat{f}$:

$$f(x) - \inf_{A(x+v)=b} \hat{f}(x+v) = \frac{1}{2}\lambda(x)^2$$

$$\Updownarrow$$

$$f(x) - \hat{f}(x + \triangle x_{nt}) = - \bigtriangledown f(x)^T \triangle x_{nt} - \frac{1}{2} \triangle x_{nt}^T \bigtriangledown^2 f(x) \triangle x_{nt}$$

- directional derivative in Newton direction:

$$\left.\frac{d}{dt} f(x + t\Delta x_{\mathrm{nt}})\right|_{t=0} = -\lambda(x)^2$$

$$\left(\left.\frac{d}{dt} f(x + t \triangle x_{nt})\right|_{t=0} = \frac{f(x + t \triangle x_{nt}) - f(x)}{t} = \bigtriangledown f(x)^T \triangle x_{nt} = -\lambda(x)^2\right)$$

- in general, $\lambda(x) \neq \left(\nabla f(x)^T \nabla^2 f(x)^{-1} \nabla f(x)\right)^{1/2}$

Newton's method with equality constraints

given starting point $x \in \mathbf{dom}\, f$ with $Ax = b$, tolerance $\epsilon > 0$.

repeat

1. Compute the Newton step and decrement Δx_{nt}, $\lambda(x)$.
2. *Stopping criterion.* **quit** if $\lambda^2/2 \leq \epsilon$.
3. *Line search.* Choose step size t by backtracking line search.
4. *Update.* $x := x + t\Delta x_{\mathrm{nt}}$.

- a feasible descent method: $x^{(k)}$ feasible and $f(x^{(k+1)}) < f(x^{(k)})$

Newton's method and elimination

- Newton's method with equality constraints is equivalent to Newton's method to solve the unconstrained reduced/eliminated problem:

$$\text{minimize} \quad \tilde{f}(z) = f(Fz + \hat{x})$$

  - variables $z \in \mathbf{R}^{n-p}$
  - $\hat{x}$ satisfies $A\hat{x} = b$; $\mathbf{rank}\, F = n - p$ and $AF = 0$

    - Newton's method for $\tilde{f}$: started at $z^{(0)}$, generates iterates $z^{(k)}$
    - Newton's method with equality constraints:

      when started at $x^{(0)} = Fz^{(0)} + \hat{x}$, iterates are $x^{(k)} = Fz^{(k)} + \hat{x}$

- Therefore, convergence performance is exactly like the performance of Newton's method to solve unconstrained problems

Infeasible start Newton method

- A generalization that deals with infeasible initial points and iterates
  - let x be a point that we do not assume to be feasible
  - find $x + \triangle x_{nt}$ that solves the second-order approximation
    (see also slide 10)

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x_{\mathrm{nt}} \\ w \end{bmatrix} = - \begin{bmatrix} \nabla f(x) \\ Ax - b \end{bmatrix}$$

  - The Newton method is based on primal-dual Newton step: both primal variable x and dual variable ν are updated

Infeasible start Newton method

- Primal-dual Newton step

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x_{\text{nt}} \\ \nu^+ \end{bmatrix} = - \begin{bmatrix} \nabla f(x) \\ Ax - b \end{bmatrix}$$

where $\nu^+ = \nu + \triangle \nu_{nt}$ (equivalent to w on the previous slide)

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x_{\text{nt}} \\ \Delta \nu_{\text{nt}} \end{bmatrix} = - \begin{bmatrix} \nabla f(x) + A^T \nu \\ Ax - b \end{bmatrix}$$

primal Newton step

dual Newton step

- Residuals

$$\begin{aligned} r(x, \nu) &= (\nabla f(x) + A^T \nu, Ax - b) \\ &= (r_{dual}(x, \nu), r_{pri}(x, \nu)) \end{aligned}$$

optimality condition <=> $r(x, \nu) = 0$

Infeasible start Newton method

- Primal-dual Newton step (an alternative way to derive)
  - write optimality condition as $r(y) = 0$, where

$$y = (x, \nu), \qquad r(y) = (\nabla f(x) + A^T \nu, Ax - b)$$

can be understood as $(r_{dual}(x, \nu), r_{pri}(x, \nu))$

  - linearizing $r(y) = 0$ gives $r(y + \Delta y) \approx r(y) + Dr(y)\Delta y = 0$

Derivative of r evaluated at y

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x_{\rm nt} \\ \Delta \nu_{\rm nt} \end{bmatrix} = - \begin{bmatrix} \nabla f(x) + A^T \nu \\ Ax - b \end{bmatrix}$$

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x_{\rm nt} \\ \nu^+ \end{bmatrix} = - \begin{bmatrix} \nabla f(x) \\ Ax - b \end{bmatrix}$$

where $\nu^+ = \nu + \triangle \nu_{nt}$

Infeasible start Newton method

given starting point $x \in \mathbf{dom}\, f$, ν, tolerance $\epsilon > 0$, $\alpha \in (0, 1/2)$, $\beta \in (0, 1)$.

repeat

1. Compute primal and dual Newton steps $\Delta x_{\rm nt}$, $\Delta \nu_{\rm nt}$.
2. *Backtracking line search on* $\|r\|_2$.

   $t := 1$.

   while $\|r(x + t\Delta x_{\rm nt}, \nu + t\Delta \nu_{\rm nt})\|_2 > (1 - \alpha t)\|r(x, \nu)\|_2$, $t := \beta t$.
3. *Update.* $x := x + t\Delta x_{\rm nt}$, $\nu := \nu + t\Delta \nu_{\rm nt}$.

until $Ax = b$ and $\|r(x, \nu)\|_2 \leq \epsilon$.

Infeasible start Newton method

- not a descent method: $f(x^{(k+1)}) > f(x^{(k)})$ is possible

$$\frac{d}{dt} f(x + t \triangle x_{nt})\Big|_{t=0} = \bigtriangledown f(x)^T \triangle x_{nt}$$

$$= - \triangle x_{nt}^T (\bigtriangledown^2 f(x) \triangle x_{nt} + A^T w)$$

$$= - \triangle x_{nt}^T \bigtriangledown^2 f(x) \triangle x_{nt} + (Ax - b)^T w$$

- the norm of r decreases in the Newton's direction:

let $y = (x, \nu)$, $\quad \frac{d}{dt} \|r(y + t\Delta y)\|_2\Big|_{t=0} = -\|r(y)\|_2$

$$\frac{d}{dt}||r(y + t \triangle y)||_2^2\Big|_{t=0} = \frac{d}{dt} r(y + t \triangle y)^T r(y + t \triangle y)\Big|_{t=0} = 2r(y + t \triangle y)^T \frac{dr(y + t \triangle y)}{dt}\Big|_{t=0}$$

$$= 2r(y)^T Dr(y) \triangle y = -2r(y)^T r(y)$$

taking derivative of square, we have $2||r(y + t \triangle y)||_2 \frac{d||r(y + t \triangle y)||_2}{dt}\Big|_{t=0} = -2||r(y)||_2^2$

- if t=1, the next iterate will be feasible, and all the following iterates will be feasible

Example

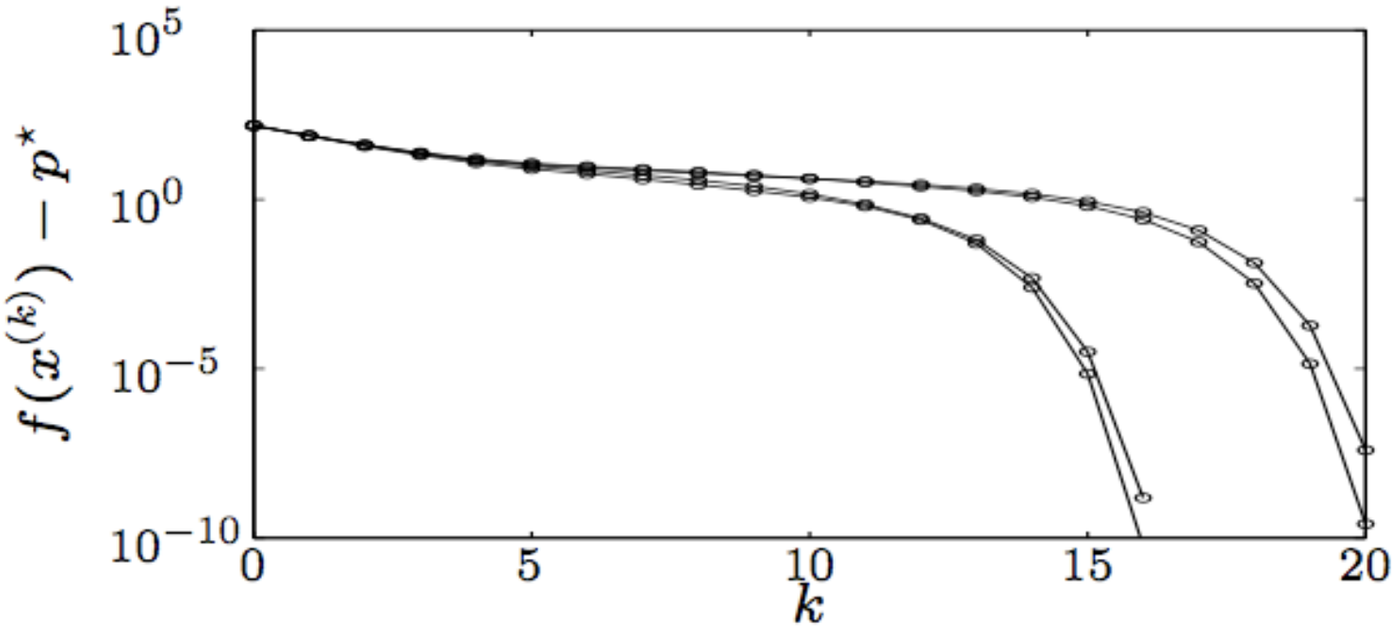
- Equality constraint analytic centering

  primal problem: minimize $-\sum_{i=1}^{n} \log x_i$ subject to $Ax = b$

  dual problem: maximize $-b^T\nu + \sum_{i=1}^{n} \log(A^T\nu)_i + n$

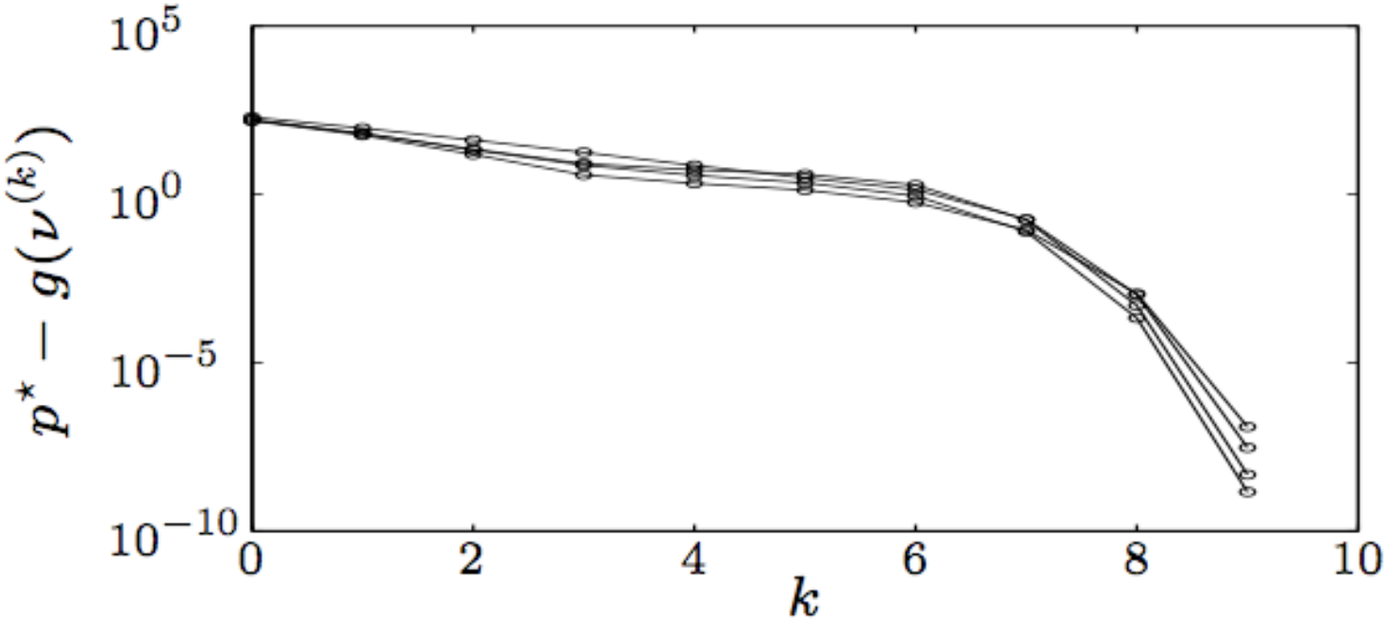
three methods for an example with $A \in \mathbf{R}^{100\times 500}$, different starting points

1. Newton method with equality constraints (requires $x^{(0)} \succ 0$, $Ax^{(0)} = b$)

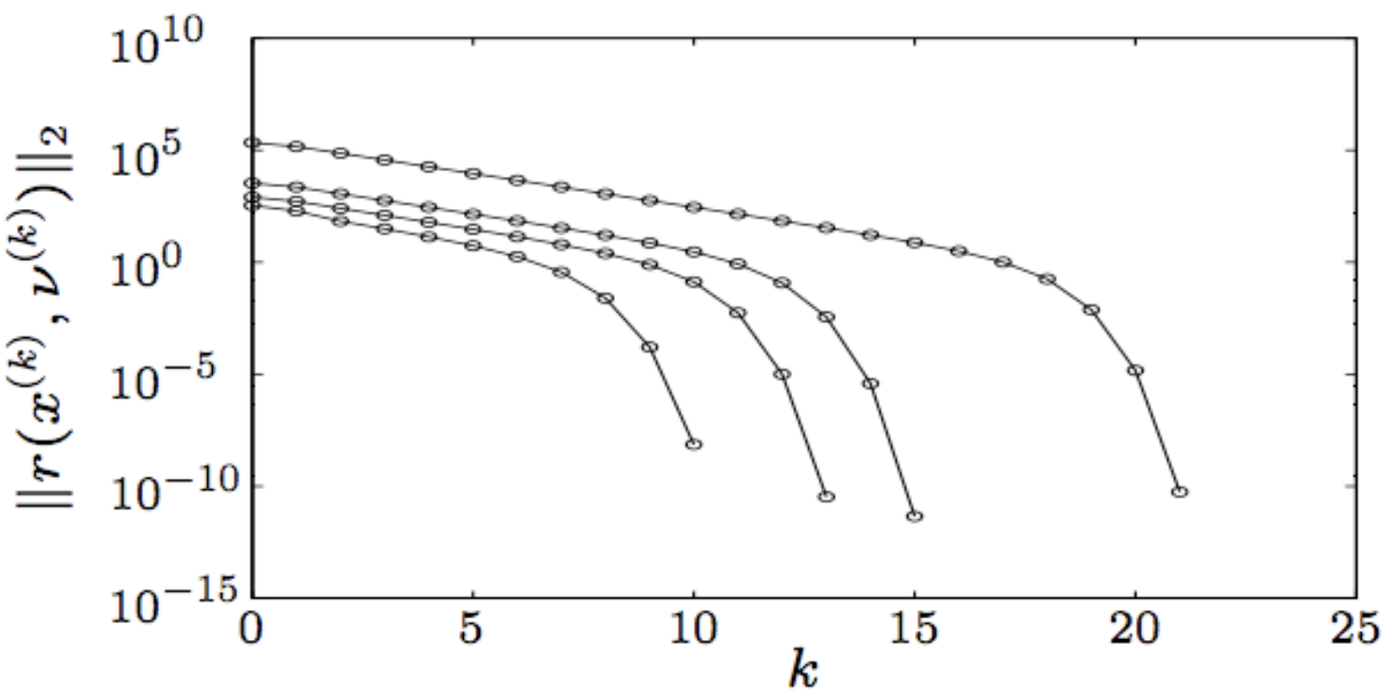


Example

2. Newton method applied to dual problem (requires $A^T\nu^{(0)} \succ 0$)



3. infeasible start Newton method (requires $x^{(0)} \succ 0$)



- Reference
  - Chapter 10, Convex Optimization.
- Acknowledgement
  - Some materials are extracted from the slides of Stanford EE364a and EE364b